

The Solow Paradox of Generative AI

Wages Flat, Postings Down, the Commons Hollowing Out

Audit synthesis

Humlum & Vestergaard (2025), *BFI WP 2025-56* | Liu, X. (2025), *Working Paper*
Burtch, Lee & Chen (2024), *Scientific Reports* 14:10413 | Lyu et al. (2025), *WWW Companion '25*
(Acemoglu et al.)

“You can see the computer age everywhere
but in the productivity statistics.”

— Solow (1987), invoked by Humlum (2025) at p. 22.

0% | -19% | -12% | -29,013

wage effect on incumbents | entry postings (Y3) | Stack Overflow traffic |
Wikipedia views/article

The aggregates show nothing. The pipeline and the commons say
otherwise.

Three places to look for displacement — only one shows zero

Where to look	What it measures	Signal
Incumbent wages & hours	Pass-through of productivity to pay	Zero (Humlum 2025)
New-hire demand	Postings, hires, vacancy mix	-12% to -19% (Liu 2025)
Knowledge commons	Q&A traffic, edits, contributions	Hollowing (Burtch 24; Lyu 25)

Stocks (incumbents) look fine. Flows (entry, contribution) do not.

Incumbents look untouched

Adoption is wide. Pass-through is narrow.

Adoption ran from 47% to 83%. Pass-through to earnings was 3–7%.

Margin	No encourage.	Encouraged
“Ever used” AI chatbot	47%	83%
Daily use	8%	21%
Time savings (% of work hours)	~2.0%	2.8–6.8%
Pass-through to earnings	~3%	~7%

Workers *used* the tool. Time *was* saved. Almost none of it landed in pay.

Humlum & Vestergaard (2025), Tables 1–2; Figs. 2, 4, 10. $N \approx 25,000$ workers, 11 occupations.

Across 11 exposed occupations, the dynamic DiD rules out wage effects above 1%

Specification	CI on % wage effect	F or N
Dynamic DiD (Fig. 7)	rules out >2%	N $\approx$ 25K
Pooled DiD “All” (Fig. 8a)	rules out >1%	11 occs
Per-occupation pooled	rules out >6%	each
Coworker IV (Tab. E.1)	confirms OLS	F = 3,645
Workplace-level DiD (Fig. 11)	rules out >1%	wage bill

Same answer in five different cuts. Adopters, encouraged adopters, daily users, early adopters, and high-adoption workplaces are all statistically indistinguishable from controls.

Humlum & Vestergaard (2025), Figs. 7–11; admin data through June 2024.

But the door is closing on new hires

What incumbents kept, entrants will not get.

Job postings in AI-substitutable occupations fell 12% on average — and the effect is widening

Horizon (post-ChatGPT)	Δ postings	SE / sig.
Pooled average	−12.2%	0.036 (***)
Year 1 (2023)	−6.4%	~0.036 (*)
Year 2 (2024)	−13.5%	~0.036 (**)
Year 3 (2025)	−18.2%	0.036 (***)

Decile $\times$ quarter FEs absorb exposure level; identified margin is *substitutability*.
Pre-trend Wald: $F = 0.83$, $p = 0.66$.

Liu (2025), Table 1, Col. 3; 285M Lightcast postings, $N = 6,753,930$ cells, 2018Q1–2025Q2.

Entry-level and administrative postings absorb the entire decline

Slice	Δ postings	Note
Entry-level	−19%	largest age effect
Mid-level	smaller	attenuates
Senior	near zero	insulated
Administrative support	−40%	sectoral peak
Professional services	−30%	second largest

The earnings of incumbent senior workers are stable. The *path into* those jobs is not.

Liu (2025), heterogeneity analysis; experience and 2-digit industry interactions.

And the knowledge commons hollows out

The training data degrades when its contributors leave.

Stack Overflow lost $\sim 1\text{M}$ visitors per day; Reddit's developer subs lost zero

Outcome	Stack Overflow	Reddit dev subs
Daily traffic (SCUL, Fig. 1)	-12% ($\sim 1\text{M}/\text{day}$)	null
Question volume (DiD, Fig. 2)	-10 to $-15/\text{wk}$	null
Hardest-hit topics (SO)	Python, JS, ReactJS, Pandas	n/a

Social fabric moderates. LLMs substitute for pure information; communities with bonds (Reddit) absorb the shock, communities without them (Stack Overflow) bleed.

Burtch, Lee & Chen (2024), Figs. 1–4; SimilarWeb daily, Sept 2022–Mar 2023.

Newer users leave first; the questions that remain are harder

Compositional shift, week 15 post-ChatGPT	Magnitude
Avg. posting-user account tenure	+100 to +200 days
Avg. long-word frequency in questions	+0.3 to +0.5 units

The juniors who left were learning by asking. They lose mentorship and peer feedback — and they are precisely the cohort least able to verify a hallucinated LLM answer.

Burtch, Lee & Chen (2024), Figs. 5–6; replicated via Athey et al. (2021) matrix completion in Fig. S2.

Wikipedia articles substitutable by GPT lose 29,013 views — and the edits drop too

Outcome	DiD coefficient	Sig.
Pageviews / article-month	−29,013 (SE 6,904)	$p < 0.01$
Edits / article-month	−6.88 (SE 3.72)	$p < 0.10$

Treatment = above-median cosine similarity to GPT-3.5 output. $N=2,206$ articles, 24-month event window. The view drop is robust; the edit drop is suggestive but only marginally so.

The same loop as Stack Overflow. Fewer readers $\Rightarrow$ fewer editors $\Rightarrow$ weaker training corpus.

Lyu et al. (2025), *WWW Companion '25*, Table 1; Acemoglu among co-authors.

Devil's advocate: aren't these all just the J-curve before the J turns up?

The strongest objection

Brynjolfsson, Rock & Syverson (2021): firms in the trough of a productivity J-curve sink complementary investments now that pay off later. Humlum (2025, p. 21) flags this; his most common new task is “AI integration” itself.

Reply. J-curve explains the zero on incumbent wages — not:

- Liu (2025): entry postings declined three years running (-6.4 , -13.5 , -18.2%). A J-curve trough does not deepen monotonically.
- Burtch (2024): contributor exit is *compositional*; newer users leave. That selection cannot be reversed by waiting.
- Lyu (2025): the commons that trains the models is meanwhile shrinking.

J-curve is incumbent-side optimism. The cost falls on entrants and on commons.

Stocks are stable. Flows are bleeding. The signal hides in the latter.

Variable type	Signal	Empirical anchor
Incumbent stock	precise zero	Humlum: $CI < 1\%$ on wages
Entry flow	−19% Y3 entry	Liu: $-6.4 \rightarrow -18.2\%$ over 3 yrs
Knowledge inflow	−12% traffic	Burtch: SCUL on Stack Overflow
Knowledge upkeep	−29K views	Lyu: DiD on Wikipedia

The economic measurements that would call this an “effect” — earnings, hours, employment of incumbents — are exactly the variables that miss it. Look at flows.

The aggregates are clean.

The pipeline and the commons are not.